

Introduction to Euro-Q-Exa (IQM @ LRZ)

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LRZ Friendly User Pilot Phase Kickoff

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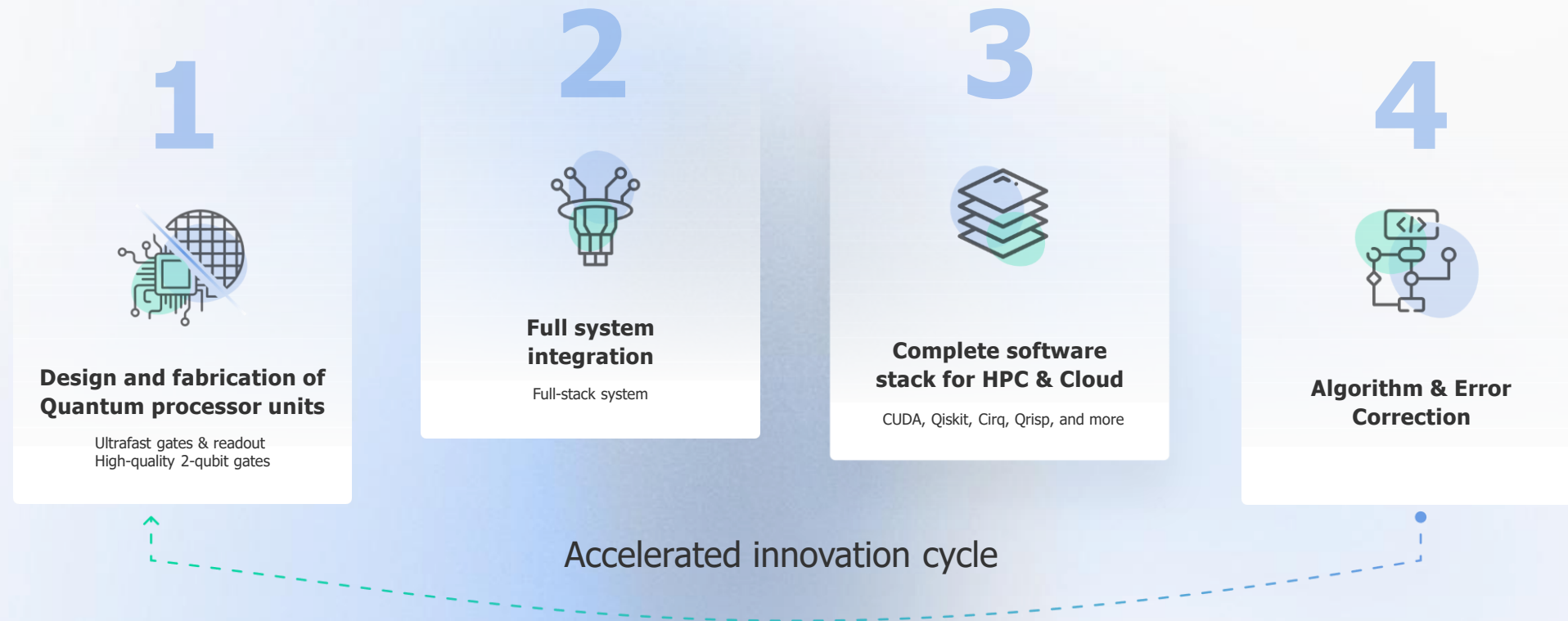
www.meetiqm.com



We build world leading
quantum computers for
the well-being of
humankind, now and
for the future



We build full stack superconducting qubit quantum computers

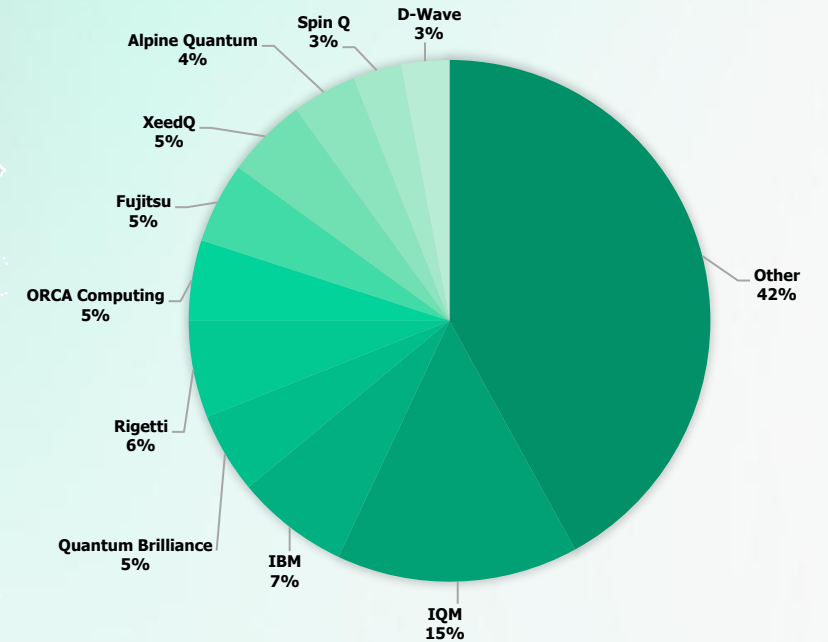


IQM Customer Deliveries

Many of our customers are leading HPC centers



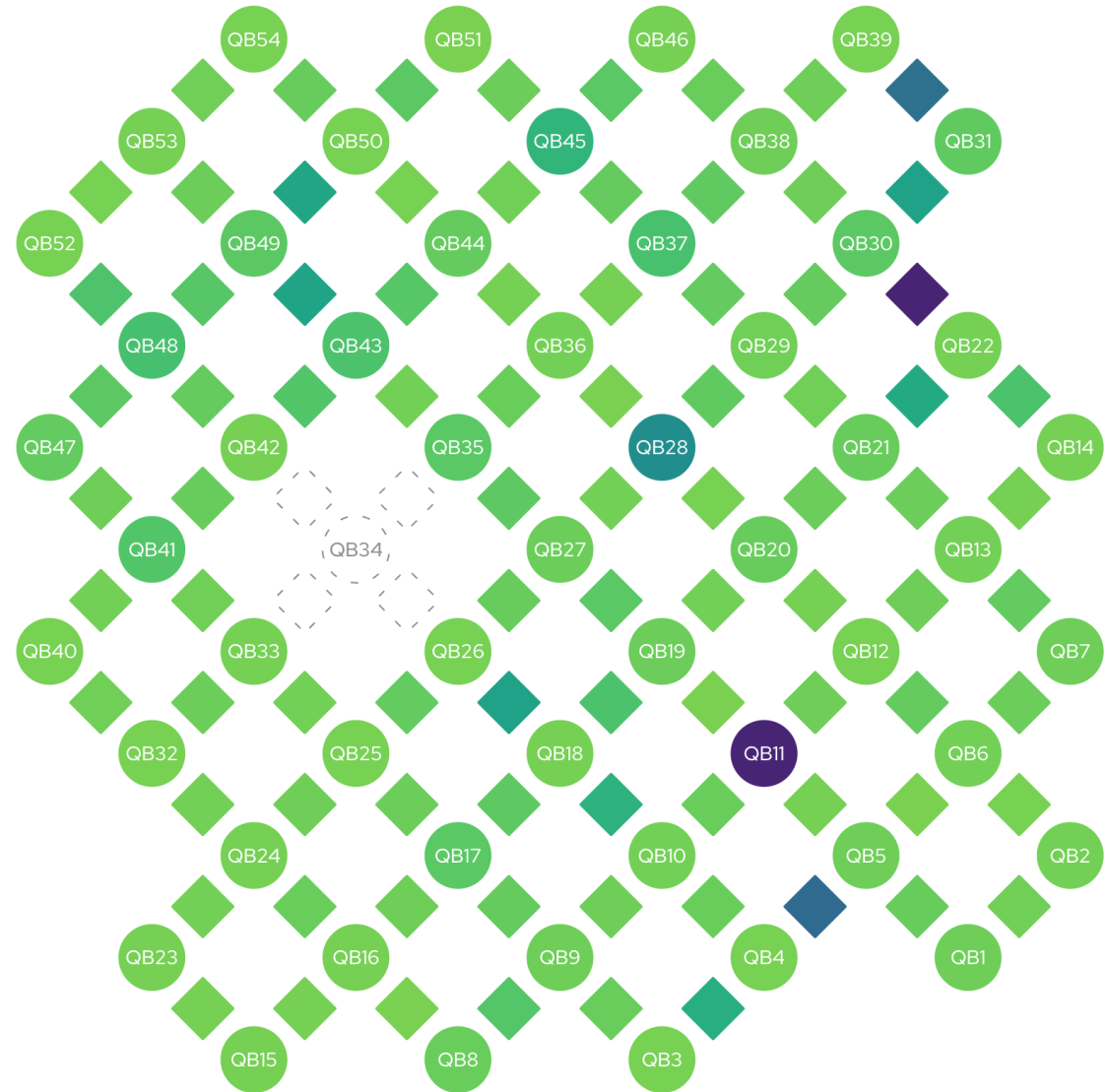
Top-10 vendors by share of total systems sold, 2020 to June 2025



Euro-Q-Exa

At a glance

Qubit type	Superconducting
Qubit count	53
Connectivity	Square lattice
Native gate set	Rx, Ry, CZ



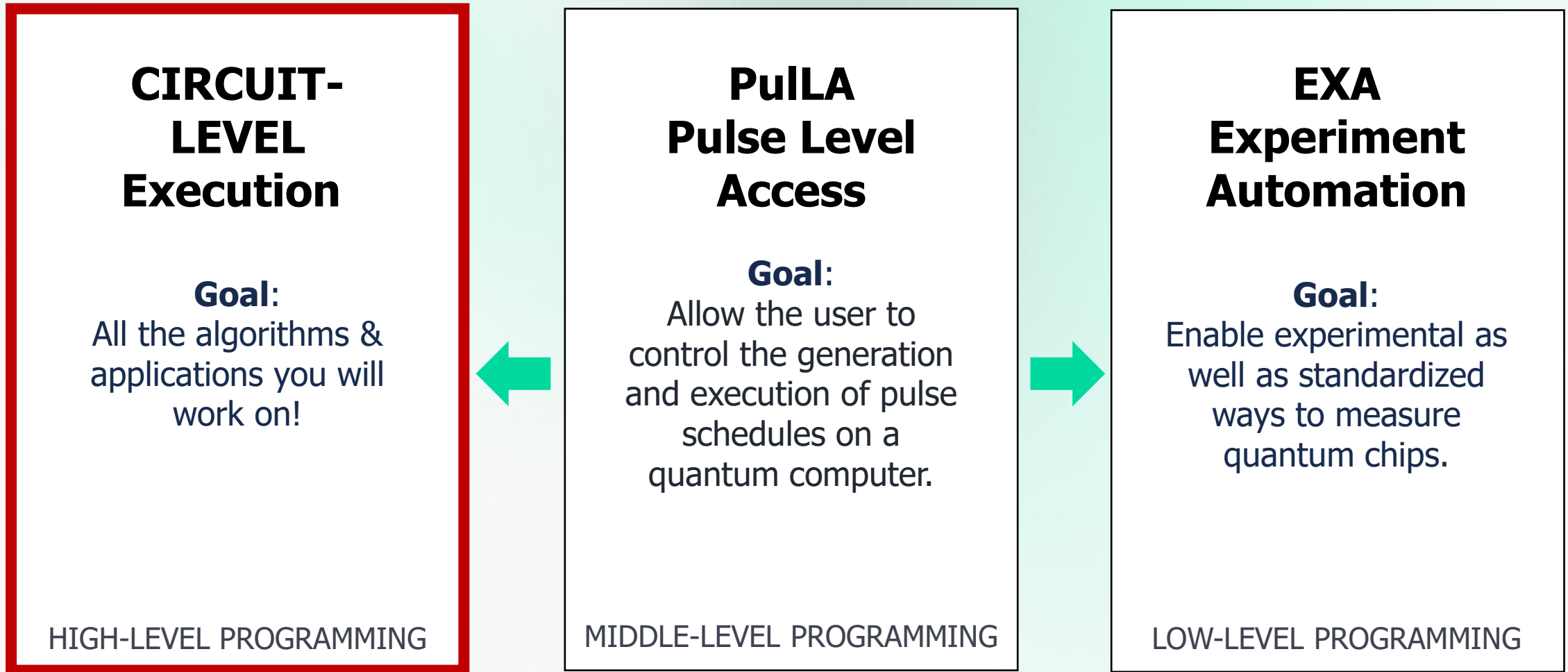
— Tech specs (as of March 9 calibration)

Coherence times	
Median T1	44.77 μs
Median T2 (echo)	45.75 μs
Median T2 (Ramsey)	7.09 μs

Gate fidelities	
Median PRX fidelity	99.96%
Median CZ fidelity	98.84%

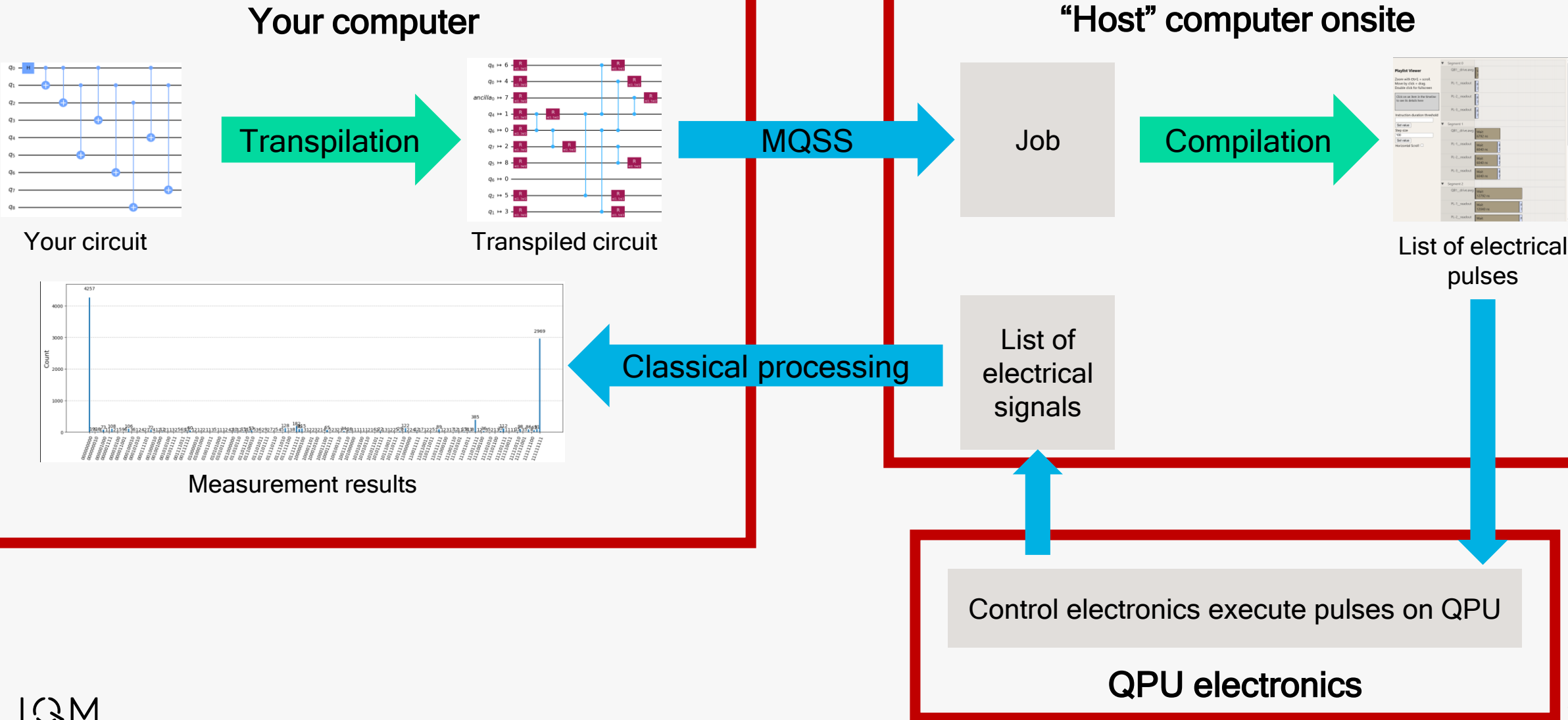
(These numbers fluctuate slightly on a couple of days time scale.)

Capabilities

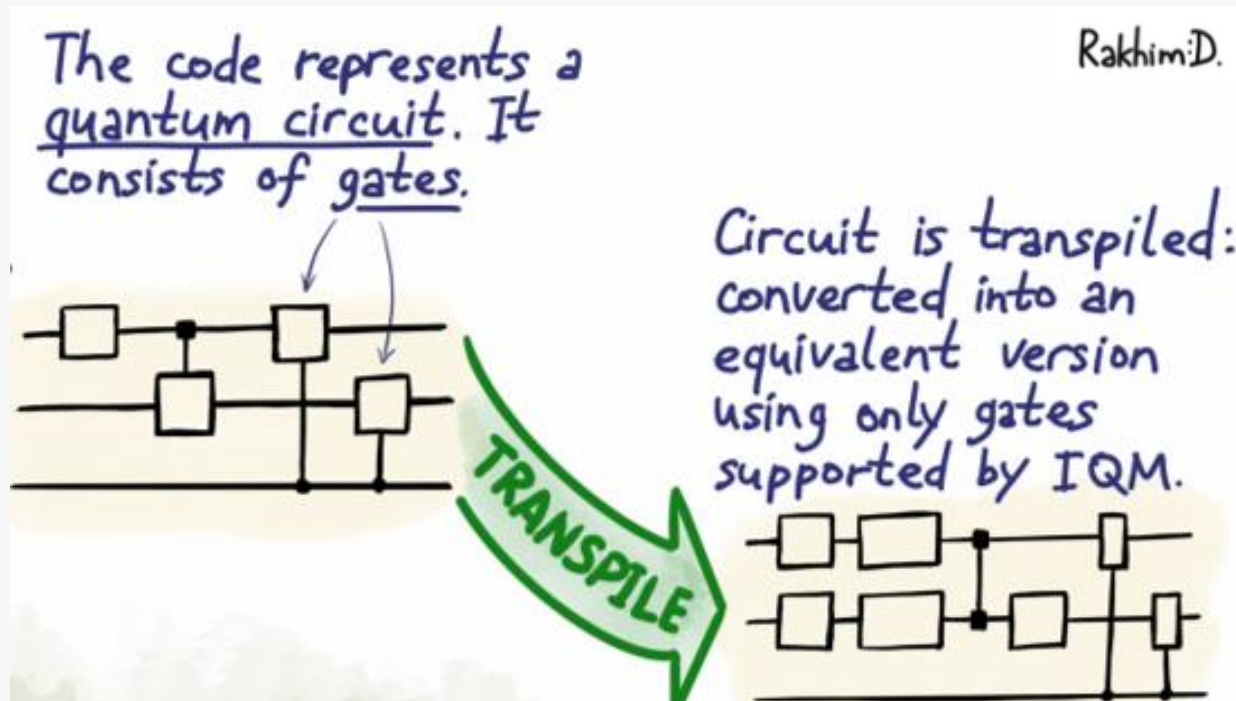


**What happens
when you run a
circuit?**

Overall flow



Transpilation



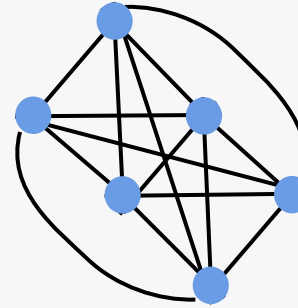
Transpiling describes the process of converting a quantum circuit into an equivalent quantum circuit that is compatible with a specific quantum hardware's constraints and gate set.

Transpilation-aware circuit design can avoid noise!

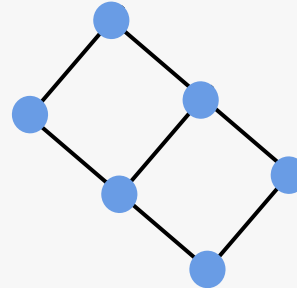
Circuit routing

Routing (= solving QPU connectivity constraints)

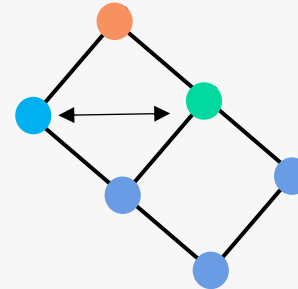
- Quantum algorithms assume arbitrary, even all-to-all connectivity
- Connectivities of QPU's are usually limited to nearest neighbor graphs
- Solution: dynamically remap **logical qubits** ("qubit 1" in the program") to the graph of **physical qubits** ("qubit 1" on the physical hardware) with SWAP gates



Algorithm's
connectivity



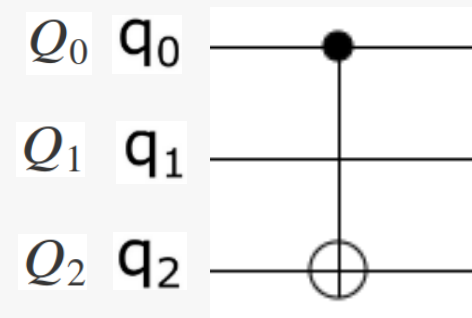
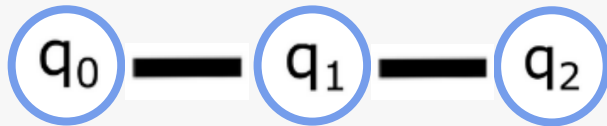
QPU's
connectivity



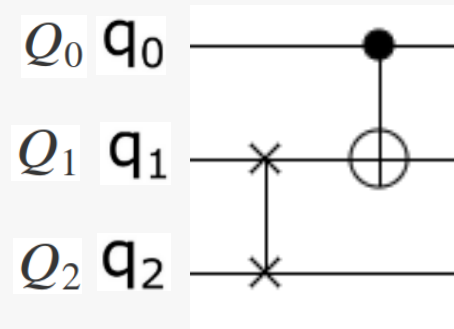
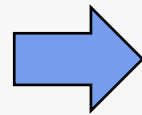
Dynamical
remapping

Routing example

Consider the following connectivity:



Routing incompatible operation

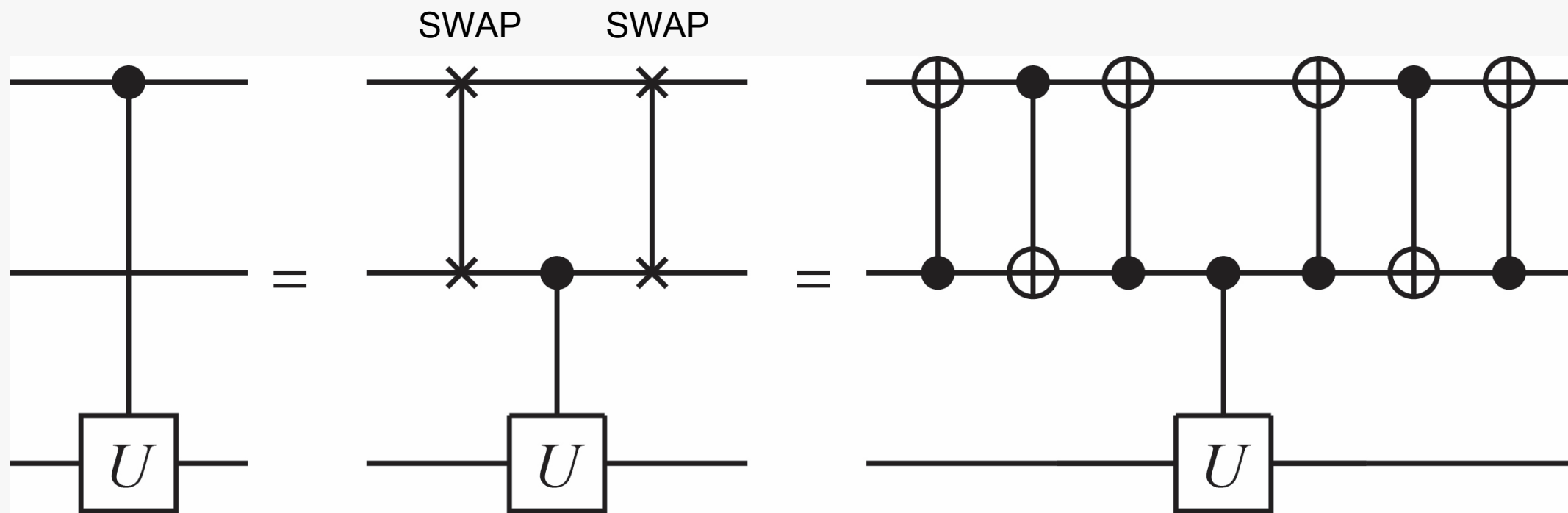


Routing compatible operation

$$\text{SWAP} = \begin{array}{c} \text{X} \\ \text{X} \end{array} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

SWAP gate swaps the states of two qubits

SWAP is expensive!



Gate decomposition

Implementing gates on a QPU

- We need to be able to implement *any* unitary operation on our quantum computer. But not all operations are simple on all hardware.
- The set of simple gates used on a *specific* quantum device is called the **native** gate set of the device.
 - All operations must be rewritten as a sequence of native gates.

IQM systems support arbitrary X and Y rotations as the native single-qubit gates and CZ as the native two-qubit gate.

— Universal gate set

- We need to be able to implement *any* unitary operation on our quantum computer. But we need enough native gates to do that.
- Our native gate set should be **universal**. What does that mean?

A set of gates S is said to be **universal** if **any** unitary operation can be approximated to any desired precision by composing only the gates in the set S .

— Universal gate set

A universal set of gates for quantum computing can:

- Create superposition of states
- Create entangled states
- Contains gates with real AND complex entries
- Contains more than the "Clifford group"

Special quantum circuits which can be efficiently simulated by a classical computer. To read more: "Gottesman-Knill theorem"

Universal gate set

Is there only one universal set of gates? If so, we'd be forced to use hardware with a very specific set of native gates...

No, there exist different universal gate sets, such as:

$$\{CNOT, H, S, T\}$$

$S = \text{phase gate}$

$T = \pi/8 \text{ gate}$

$$\{CZ, R_x(\theta), R_y(\phi)\}$$

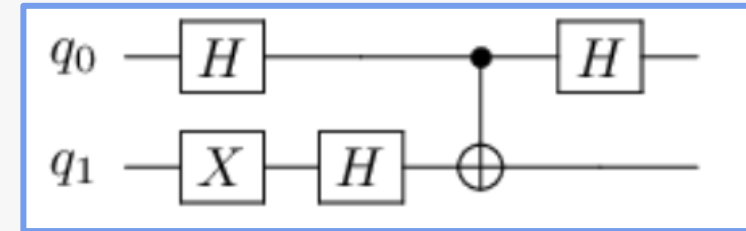
$$\{H, TOFF\}$$

$TOFF = \text{"Toffoli gate"}$

Decomposing (= writing a gate a series of native gates)

- Circuits of quantum algorithms usually use 'intuitive' gates like CNOT, X, H, P, SWAP
- Real QPU's use only few *native*, hardware friendly gates to process their qubits
- Solution: gate identities can convert the algorithm's circuit to equivalent circuit suitable for given QPU

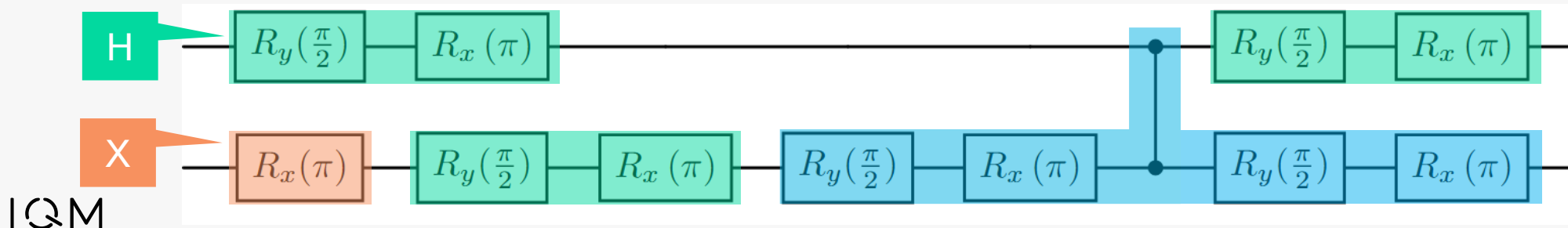
Circuit for Deutsch's algorithm with CNOT as the oracle



Example QPU native gates

$$\{R_x(\theta), R_y(\phi), CZ\}$$

CNOT



Redefining more precisely...

Transpiling = routing circuit to
connectivity of a QPU and
decomposing gates to its
native gates

A little bit about compilation

After transpilation:

- Your request is compiled by a server into a sequence of pulses



Dilution refrigerator

Electronics rack

After transpilation:

- Your request is compiled by a server into a sequence of pulses
- Control and readout pulses are generated in the electronics rack and sent into the refrigerator



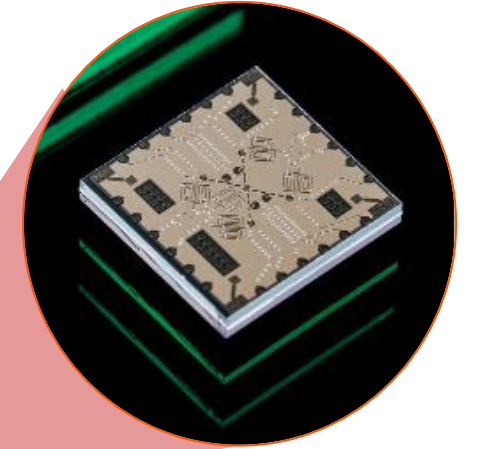
Dilution refrigerator

Electronics rack

After transpilation:

- Your request is compiled by a server into a sequence of pulses
- Control and readout pulses are generated in the electronics rack and sent into the refrigerator
- The pulses interact with the QPU

After that, the measurement device “reads” the results, and you can analyze them on your PC.



Dilution refrigerator

Electronics rack

Standard stages of compilation in IQM devices

CIRCUIT VALIDATION

If no custom implementation is specified, the default implementation for each gate are selected automatically.

CIRCUIT RESOLUTION

The circuit is converted to *Timeboxes*: containers for instruction schedule fragments, to be scheduled according to a given scheduling strategy.

TIMEBOX LEVEL

The timeboxes corresponding to the measurement instructions in the circuit are added.

TIMEBOX RESOLUTION

All relative timings between pulses are resolved and fixed. The different timeboxes are assembled in a single schedule.

Standard stages of compilation in IQM devices

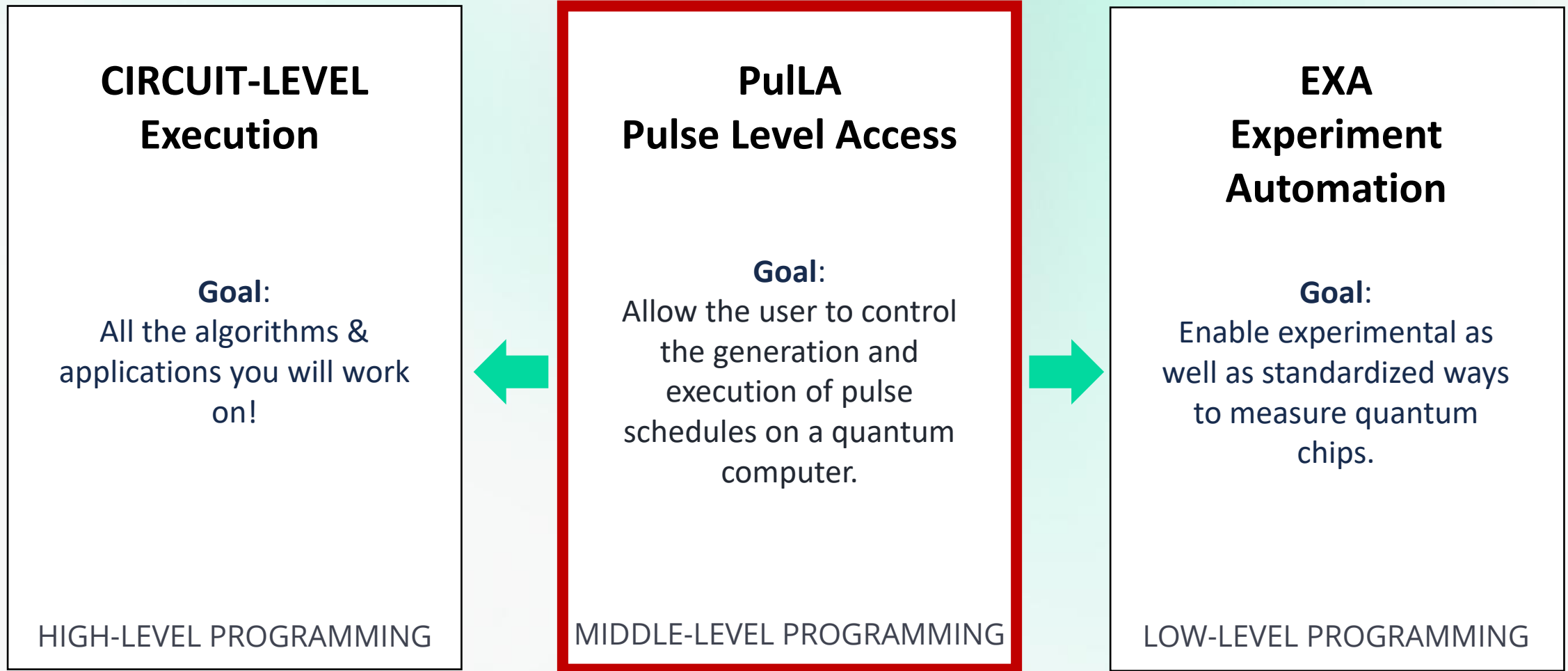
After these main compilation stages, before executing the circuit, one can do an extra pass if wanted/needed:

DYNAMICAL DECOUPLING

When the circuit calls for a qubit to do nothing, special extra pulses are inserted in order to **suppress errors**.

IQM software supports a flag for enabling dynamical decoupling when running circuits
– this should be supported soon via MQSS access.

Information about compiled circuits is available in PuLLA



Resources

IQM Academy

Your one-stop shop for online quantum education

Gain a thorough understanding of quantum computing and its applications.

- A free curriculum and learning platform for anyone
- Easy to pick up, no previous knowledge required
- Highly interactive and engaging

<https://academy.meetiqm.com/>

IQM Community Discord

Connect with the IQM community

Gain a thorough understanding of quantum computing and its applications.

- Chat about your latest project
- Get help from IQMers or other IQM users
- Stay in the loop about IQM events

<https://discord.gg/F38bwqPYJ4>

IQM

IQM Docs

Your one-stop shop for support

Full documentation on capabilities and interfaces with IQM machines

- Video tutorials
- Documentation for IQM client-side libraries
- Information on pulse-level control

<https://docs.meetiqm.com/>